

CHEN HEXIN

Beijing, P.R.China

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EDUCATION

Tsinghua University, China

09/2024 - 06/2028 (Expected)

B.S. in Mathematics and Physics; B.E. / B.S. in Software Engineering (Dual Degree)

GPA: 3.8/4.0; Ranking: top 15%

Relevant Coursework: Calculus, Linear Algebra, Probability and Stochastic Progress, Data Structures, Discrete Mathematics I-II, Fundamental Programming, Principles of Database Systems, Computer Organization Principle

RESEARCH EXPERIENCE

Cognitive-Breath: Cognitive Load Sensing through Eyeglass Nose Pads

Chicago, USA

Research Assistant; Advisor: Dr. Pedro Lopes, Human Computer Integration Lab, University of Chicago

06/2026 - 09/2026

- Developed an end-to-end pipeline for cognitive-load sensing from breathing signals captured by contact microphones embedded beneath eyeglass nose pads, spanning signal preprocessing, feature engineering, model development, cross-user evaluation, and real-time inference.
- Processed acoustic signals using 150–1000 Hz Butterworth band-pass filtering, 64 ms framing, and log root-mean-square (RMS) envelope extraction; engineered temporal-spectral features capturing breathing rhythm, signal intensity, cycle morphology, and bilateral-channel synchrony.
- Built lightweight classifiers using regularized logistic regression and gradient boosting, and introduced causal online mean/variance normalization with participant-specific calibration to mitigate cross-user distribution shifts, improving best block-level classification accuracy by approximately 17% relative to the baseline.
- Conducted leave-one-participant-out cross-user evaluation, feature ablation, bilateral-channel fusion, and robustness analyses under acoustic and motion interference; also identified sampling-rate drift in synchronized acoustic and nasal-pressure recordings to improve temporal alignment and validation.
- Implemented a streaming inference and visualization prototype that continuously processed live sensor data and updated cognitive-load estimate every 5 seconds while preserving causal, calibration-free inference for unseen users.
- Publication: Co-first-author manuscript currently under review.

Audio Signal Representation Project

Beijing, China

Research Assistant; Advisor: Dr. Jiangyan Yi, Tsinghua University

06/2025 - 03/2026

- Developed a neural audio representation pipeline for academic research and prototype-level experiments.
- Refactored the quantization module to improve modularity, reconstruction stability, and downstream usability.
- Implemented and evaluated alternative vector-quantization designs for separating global audio patterns from local acoustic details.
- Conducted controlled experiments and ablation studies to compare system components.

Motion-Robust IMU Signal Processing for Wearable Interaction
Research Assistant, Tsinghua University

Beijing, China
09/2024 - 07/2025

- Analyzed multi-axis accelerometer signals in both time and frequency domains, identifying noise patterns and non-stationary motion artifacts in fine-grained wearable sensing data.
- Developed an adaptive signal filtering pipeline to reduce motion-induced jitter while preserving subtle intentional movement cues under low-latency constraints.
- Improved trajectory stability and repeatability in prototype experiments with minimal impact on system responsiveness.

LEADERSHIP & ACTIVITIES

Group Lead, Student Practice Committee, Wuqiong College, Tsinghua University	08/2025 - Present
Team Captain, “Qinglai Zhixing” Social Practice Team, Tsinghua University	07/2025 - 08/2025
Participant, MCM/ICM Mathematical Modeling Contest	02/2025
Team Lead, Tsinghua - CUC Game Jam	12/2024

HONORS & AWARDS

Silver Award, Summer Social Practice Team, Tsinghua University	12/2025
Academic Excellence Scholarship	10/2025
Merit Scholarship	10/2025
Successful Participant Award, MCM/ICM Mathematical Modeling Contest	01/2025

CORE SKILLS

Languages: Mandarin Chinese (Native); English (TOEFL 100)

Mathematics: Calculus, Linear Algebra, Probability & Stochastic Processes, Discrete Mathematics, Mathematical Modeling, Statistical Analysis

Programming & Engineering: Python, C++, C, Java, Arduino; Signal Processing, Databases, Computer Systems, Software Engineering

Machine Learning & Research: Machine Learning, Time-Series Analysis, Feature Engineering, Model Training & Evaluation, Controlled Experiments, Statistical Analysis